

	Task	Document	Examples	Role	ER@1	SR@1
1	✓				16.00	25.00
2		✓			42.00	47.62
3	✓	✓			84.00	71.43
4			✓		92.00	73.91
5	✓		✓		92.00	76.09
6		✓	✓		92.00	76.09
7	✓	✓	✓		92.00	78.26
8		✓	✓	✓	94.00	78.72
9	✓	✓	✓	✓	94.00	80.85

Table 5: Results of different prompt components for tree asset generation. Examples, Role, Document and Task represent four integral components of the prompt template.



Figure 7: Visualization of the personalized editing results on different scenes.

eration and accelerating the industrial production process.

Ablation Study

To analyze the impact of various components within the systematic template, we conduct an ablation study based on the tree plugin in PCGHub. During the experiment, we utilize the dataset from Section 4.1 for testing, maintaining a consistent format. We incrementally add or remove different parts of the systematic template, using ER@1 and SR@1 as metrics to observe the impact of various components on the system. The results are shown in Table 5. It is evident from the results that augmenting the Task, Document, Examples and Role components contributes to an increase in ER@1 and SR@1. Among these, the inclusion of the Example component results in the most significant improvement, resulting in a maximum increase of 76.00% and 51.09% in ER@1 and SR@1 respectively. Conversely, the Role component has the least impact, with maximum increases of 2.00% and 2.59% in ER@1 and SR@1 after its addition. These results suggest that the setting of Task significantly impacts the performance of LLMs. By clarifying the goals and requirements of the task, the system can better understand the user’s intent and better meet the needs when generating output. Examples play an important role in designing proxy prompts. By providing concrete examples, the system can better understand the user’s intent and needs and produce high-quality output related to the input text. Documentation provides background information that can help the system better meet user expectations when generating output.

Comparing with Different LLMs

To investigate the performance of different variants of large language models (LLM) in Scene $\mathcal{X}$, we test public LLM

Model	Scene Generation		Asset Generation	
	ER@1	SR@1	ER@1	SR@1
Llama2-7B(Touvron et al. 2023)	30.00	53.33	38.00	57.89
Llama2-13B(Touvron et al. 2023)	44.00	59.09	54.00	66.66
Mistral(Jiang et al. 2023a)	76.00	68.42	94.00	85.11
Gemma-2B(Team et al. 2024)	6.00	33.33	22.00	45.45
Gemma-7B(Team et al. 2024)	36.00	55.56	68.00	73.53
GPT-3.5-turbo(Brown et al. 2020)	66.00	60.60	82.00	82.93
GPT-4(Achiam et al. 2023b)	86.00	86.05	96.00	85.42

Table 6: Comparing the performance of different language models in natural scene generation and city generation.

APIs like gpt-3.5-turbo, gpt-4 and several external open-source LLMs in this subsection. To ensure the stability of LLM outputs, we set the temperature of LLM to 0 for all experiments. We conduct experiments on 3D scene and asset generation based on 50 scenario descriptions and 50 object descriptions in Sec. 4.1. The results are presented in Table 6. It is evident that gpt-4 delivers the superior performance, with Mistral closely following as the second-best. Due to its performance and lower hardware requirements, the open-source Mistral is a highly appealing option. When compared to asset generation, the executability and success rates noticeably decline during the generation of large-scale natural scenes, a trend that can be attributed to the increased task complexity. In particular, as the number of components involved in the system expands, the LLM may face challenges in maintaining their accuracy. Nonetheless, our method exhibits consistent performance across different LLMs, maintaining high levels of executability and success rates.

Conclusion

In this paper, we introduce Scene $\mathcal{X}$, an advanced framework for large-scale scene generation that automatically creates high-quality procedural models from textual descriptions. The framework consists of two key components: PCGHub and PCGPlanner. PCGHub includes an extensive repository of accessible procedural assets and thousands of meticulously crafted API documents. Meanwhile, the PCGPlanner can generate executable actions for Blender to create controllable and precise 3D assets following user instructions. Scene $\mathcal{X}$ can generate a 2.5 km $\times$ 2.5 km city with accurate layout and geometric structures, dramatically reducing production time from two weeks of professional work to just a few hours for an ordinary user. Our method demonstrates significant capabilities in controllable scene generation and editing, validated through extensive experiments.

Limitations. Although Scene $\mathcal{X}$ effectively enables LLM-driven procedural large-scale scene generation, it has certain limitations: 1) Dependence on the pretrained LLM model may restrict its generalization across applications. 2) It is constrained by the inherent algorithms and rules of existing PCG modules. Expanding PCGHub’s resources could further enhance scene diversity and quality.

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